# Molecular hydrogen (H2) in human health: systematized review of cellular mechanisms and clinical evidence (2007–2026, with focus on 2021–2026)

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Abstract

**Context: Molecular hydrogen (H2) has been proposed as a selective modulator of reactive species and Keap1–Nrf2 activator, with commercial expansion outpacing clinical evidence. | Objective: To synthesize cellular mechanisms and clinical evidence of molecular hydrogen (H2) in humans between 2007–2026, with focus on 2021–2026, hierarchizing by level of evidence and GRADE certainty. | Methods: Systematized review with PICO and searches in PubMed, Cochrane CENTRAL, Scopus, Web of Science, Embase, ClinicalTrials.gov and UMIN-CTR (15–20/05/2026; reproducible strings per database, Appendix B). Dual independent screening in Rayyan (agreement κ=0.86; 95%CI 0.78–0.93), standardized extraction and RoB2/ROBINS-I/AMSTAR-2. Narrative synthesis and, when homogeneous, summarization of previous meta-analyses; GRADE certainty. | Results: From 2,403 records, 82 studies were included (12 systematic reviews/meta-analyses, 34 randomized clinical trials – RCTs, n≈1,900, 18 cohorts/pilots, 18 experimental for mechanisms). H2 showed consistent safety (no serious adverse events attributed; mild events ~26% vs placebo). Benefit signals were modest and low-to-moderate certainty: total cholesterol reduction −6.7 mg/dL and LDL −3.2 mg/dL, increase in antioxidant potential (BAP) without d-ROMs reduction, discrete improvement in lower-limb explosive power and reduced perceived exertion/lactate, no effect on VO2max/strength. A phase 3 RCT (n=675) with hydrogen-rich water (HRW, 0.5–0.9 ppm, 2×/day, 21 days) in COVID-19 showed no benefit. | Conclusions: Current evidence does not support routine use of molecular hydrogen (H2). Safety profile is favorable and mechanistic plausibility exists, but observed effects are limited and based on small heterogeneous RCTs. Phase 3 RCTs with clinical outcomes and ≥12 months follow-up, with registered protocol (PROSPERO), are needed. | Keywords: molecular hydrogen; hydrogen-rich water; oxidative stress; inflammation; systematic review; GRADE**

Contents



## 1 Introduction

Since the publication of the pioneering study by Ohsawa et al. (2007), interest in molecular hydrogen ($H_2$) has grown rapidly. Although thousands of studies have been published since then, uncertainties remain regarding the consistency of the effects observed in humans and the mechanisms capable of explaining these results. The most recent evidence indicates plausible biological potential, but clinical application still depends on confirmation through multicenter trials of higher methodological quality.

In 2007, Ohsawa and colleagues described that molecular hydrogen (H2) selectively attenuates hydroxyl radicals (•OH) and peroxynitrite (ONOO−) in a cerebral ischemia-reperfusion model. Since then, thousands of publications and 81 registered trials have evaluated molecular hydrogen (H2) via oral (hydrogen-rich water, HRW, 0.5–5.9 ppm), inhalation (H2 1–66%), intravenous (hydrogen-rich saline, HRS) and topical routes, with final search date 20/05/2026.

Previous reviews (Johnsen et al., 2023, Molecules; Dhillon et al., 2024, Int J Mol Sci) mapped consistent safety but efficacy still based on small heterogeneous trials. Recent meta-analyses on lipid profile and physical performance suggest modest effects, while a phase 3 trial in COVID-19 (Hydro-COVID, n=675) showed no benefit. Mechanistically, the hypothesis of Keap1–Nrf2–ARE axis activation targeting oxidized Fe-porphyrin has gained support, but without validation in clinical outcomes.

### 1.1 Gap and objectives

Gaps remain regarding dose-response, optimal route, pharmacokinetics and clinical relevance. Commercial products containing molecular hydrogen (H2) expanded before robust evidence. This systematized review aims to synthesize cellular mechanisms and clinical evidence of molecular hydrogen (H2) in humans (2007–2026, focus 2021–2026), hierarchizing by level of evidence and GRADE certainty, following the adapted PRISMA 2020 checklist for systematized reviews.

> **PICO Objective • In adult humans, does molecular hydrogen (H2), at any quantified route and concentration, improve oxidative stress/inflammation markers and clinical outcomes compared to placebo or standard care?**

## 2 Methods

### 2.1 Protocol and registration

This is a systematized literature review (not a full systematic review in the Cochrane sense). There was no prospective PROSPERO registration — a declared limitation. For future submission as a systematic review, the protocol will be registered prospectively and the full PRISMA 2020 checklist completed (Appendix A). Reporting follows the adapted PRISMA 2020 for systematized reviews and the GRADE approach. An internal protocol with PICO, criteria and extraction plan was prepared before searches (15/05/2026).

### 2.2 Eligibility criteria

| DOMAIN | CRITERIA |
|---|---|
| Population | Human adults (≥18 years) healthy or with clinical condition (metabolic syndrome, NAFLD, T2D, CV risk, musculoskeletal condition, COVID-19, fatigue). Animal/cellular models only for mechanisms, flagged as non-extrapolatable. |
| Intervention | Molecular hydrogen (H2) at any quantified route: HRW (0.5–5.9 ppm), H2 gas 1–66% (+ O2), HRS, bath, eye drops, generating tablet. Alkaline water without H2 quantification excluded. |
| Comparator | Placebo (placebo water, ambient air), standard care or no intervention. |
| Outcomes | Primary: safety/adverse events, oxidative stress markers (d-ROMs, MDA, 8-OHdG, BAP, SOD, CAT) and inflammation (IL-6, TNF-α, CRP). Secondary: lipid/glycemic profile, hepatic steatosis, physical performance, fatigue, quality of life. |
| Study design | Included: RCTs, systematic reviews/meta-analyses, cohorts, consensus. Excluded: editorials, letters without data, uncontrolled series, non-peer-reviewed preprints, studies without comparator. |
| Other | Period 2007–2026, focus 2021–2026; languages EN/PT; peer-reviewed only. Complementary search in Google Scholar only for citation tracking. |

### 2.3 Sources of information, search strategy and selection

Searches on 15–20/05/2026 in PubMed/MEDLINE, Cochrane CENTRAL, Scopus, Web of Science Core Collection, Embase, ClinicalTrials.gov and UMIN-CTR. Google Scholar only for citation verification. No initial date filter; analytical emphasis on 2021–2026. References of reviews were hand-searched.

Two independent reviewers screened titles/abstracts in Rayyan and assessed full texts; disagreements by consensus with a third reviewer. Inter-reviewer agreement: κ=0.86 (95%CI 0.78–0.93), indicating almost perfect agreement (Landis & Koch). Database exports were archived for peer verification under the data DOI (see Declarations). The complete search strategies and exported files (.nbib, .ris, and deduplication spreadsheet) are available in the Zenodo repository.

Box 1 — Reproducible strings per database (PubMed example; adaptations in Appendix B)

| DATABASE | STRING / FILTERS |
|---|---|
| PubMed | ("hydrogen"[Mesh] OR "molecular hydrogen" OR "hydrogen-rich water" OR "HRW" OR "hydrogen-rich saline" OR "hydrogen inhalation") AND ("oxidative stress"[Mesh] OR "inflammation"[Mesh] OR "antioxidants"[Mesh]) AND ("humans"[Mesh] OR "clinical trial"[Publication Type]) NOT ("hydrogen peroxide" OR "hydrogen breath test") • Filters: 2007–2026, Humans, EN/PT |
| Scopus / Embase / WoS | TITLE-ABS-KEY(molecular hydrogen OR hydrogen-rich water OR HRW) AND TITLE-ABS-KEY(oxidative stress OR inflammation) AND TITLE-ABS-KEY(human OR clinical trial) NOT TITLE-ABS-KEY(hydrogen peroxide) • Same period |
| Cochrane | hydrogen IN Title Abstract Keyword AND (oxidative stress OR inflammation) • CENTRAL, no date limit |

| | |
|---|---|
| Registries | ClinicalTrials.gov and UMIN-CTR: "hydrogen" OR "H2" (excluding hydrogen peroxide/breath test) • Up to 09/2023 for 81 RCTs mapping |

### 2.4 Data collection and risk of bias

Standardized extraction: title, year, authors, journal, design, n, population, H2 route/concentration, duration, outcomes, results, level of evidence and limitations (Appendix C). Missing data were requested from authors when applicable.

Risk of bias: RCTs with RoB2 (domains: randomization, deviations, missing data, measurement, selection); cohorts with ROBINS-I; systematic reviews with AMSTAR-2. Independent assessment by two reviewers, with full matrix in Appendices C (34 RCTs) and D. Studies in MDPI/Frontiers were included only after verification of PROSPERO registration, PRISMA checklist and valid DOI, with sensitivity analysis excluding them.

### 2.5 Synthesis and certainty of evidence

Narrative synthesis hierarchized by outcome and, when homogeneous, summarization of previous meta-analyses (without new meta-analysis with individual data). Heterogeneity described by $I^2$ and Egger when reported. GRADE certainty per outcome (Appendix E) with downgrading for risk of bias, inconsistency, indirectness, imprecision and publication bias. No patients/public were involved in the design of this systematized review.

## 3 Results

### 3.1 Study selection

Figure 1 — PRISMA 2020 flow diagram (official model; searches 15–20/05/2026; exports archived)

| IDENTIFICATION | |
|---|---|
| Records identified from databases (n=2,403) \| PubMed 590 • Scopus 612 • Embase 384 • WoS 298 • CENTRAL 45 • Registries 81 • Other sources* 28 • Scholar 365 | Records removed before screening (n=443) \| Duplicates removed (Rayyan auto+manual) n=443 \| Records marked as ineligible by automation n=0 \| Other reasons n=0 |
| Records screened (n=1,960) \| Titles/abstracts screened n=1,960 \| Records excluded (n=1,750) | Reports sought for retrieval (n=210) \| Full texts assessed for eligibility n=210 \| Reports not retrieved n=0 |
| Reports excluded (n=128): \| • H2 without quantification 45 \| • Inadequate population 28 \| • No comparator 22 \| • No extractable data 18 \| • Duplicate 10 \| • Other 5 | Studies included in synthesis (n=82) \| • Systematic reviews/meta-analyses 12 \| • RCTs 34 (n≈1,900) \| • Cohorts/pilots 18 \| • Experimental (mechanisms) 18 |

* Other sources: references of reviews and hand search. Model adapted from Page et al., BMJ 2021.

### 3.2 Study characteristics

82 studies (2007–2026): 12 systematic reviews/meta-analyses (n=1,787 participants summarized), 34 RCTs (median n=28, range 12–675), 18 cohorts/pilots and 18 experimental studies for mechanisms. Populations: metabolic syndrome/NAFLD (9 RCTs), prediabetes/T2D (5), performance in healthy (12), aging (3), COVID-19 (2), musculoskeletal/fatigue (6). Routes with molecular hydrogen (H2): HRW 0.5–2.0 L/day 0.5–5.9 ppm (60% of RCTs), inhalation 1–66% (18%), HRS (12%), others (10%). Duration 4–24 weeks, except Hydro-COVID (3 weeks) and Zanini (24 weeks). Nine performance RCTs did not report exact concentration (limitation).

| GROUP | N STUDIES | N PARTICIPANTS | EXAMPLE / ROUTE |
|---|---|---|---|
| Reviews/Meta-analyses | 12 | 1.787 | Ye 2026, Li 2024, Zhou 2024 |
| RCTs | 34 | ~1.900 | Hydro-COVID n=675, Kura n=60 |
| Cohorts/pilots | 18 | ~400 | Zanini n=14, Korovljev n=12 |
| Mechanistic | 18 | — | Ohta 2023, Cheng 2023 |

### 3.3 Risk of bias

Of 34 RCTs, 6 (18%) low risk, 19 (56%) some concerns and 9 (26%) high risk (RoB2). Main reasons: randomization not described (n=15), deviations due to lack of blinding (n=7, inhalation), missing data >10% (n=5) and selective reporting (n=8). Cohorts (ROBINS-I) all moderate–high risk due to confounding. Reviews (AMSTAR-2): 4 of 12 moderate confidence, 8 low/critically low (lack of registered protocol and excluded list). Sensitivity excluding MDPI/Frontiers studies did not change direction but reduced heterogeneity in lipids ($I^2$ 37% → 18%). Inter-rater agreement for RoB2: κ=0.81 (95%CI 0.70–0.91).

Complete RoB2 matrix (34 RCTs) available in Supplementary Material (Table S1, Zenodo repository). Below, excerpt of the 10 main RCTs (see Supplementary Material for the 34).

| STUDY (n) | Randomization | Deviations | Missing data | Measurement | Selection | Overall |
|---|---|---|---|---|---|---|
| Hydro-COVID 2024 (675) | Low | Low | Low | Low | Low | Low |
| LeBaron 2020 (60) | Some | Low | Low | Some | Some | Some |
| Kura 2022 (60) | Some | Low | Low | Low | Some | Some |
| Korovljev 2019 (12) | High | Some | Low | Some | Some | High |
| Zanini 2021 (14) | Some | Low | Some | Low | Some | Some |
| Sládečková 2024 (12) | Some | Low | Low | Low | Low | Some |

| | | | | | | |
|---|---|---|---|---|---|---|
| Sim 2020 (20) | Low | Low | Low | Low | Some | Some |
| Ito 2024 (30) | Some | Some | Low | Some | Some | Some |
| Nakao 2023 (22) | High | Some | Some | Some | High | High |
| Mizuno 2022 (35) | Some | Low | Low | Low | Some | Some |

## 3.4 Results by outcome

Box 2 — Synthesis by outcome with GRADE certainty (full profile in Appendix E)

| OUTCOME | STUDIES / n | EFFECT (95%CI) | I² | CERTAINTY |
|---|---|---|---|---|
| Safety (serious AEs) | 34 ECRs / ~1.900 | Nenhum atribuído; EA leves 26–27% vs placebo | 0% | High |
| Total cholesterol | 13 ECRs / 757 | DM −6,71 mg/dL (−10,38 a −3,04) p<0,001 | 12% | Moderate* |
| **LDL** | 13 ECRs / 757 | DM −3,21 (−6,31 a −0,10) p=0,043 | 18% | Moderate* |
| **Triglicerídeos** | 13 / 757 | DM −4,21 (−17,75 a 9,32) p=0,54 | 0% | Low |
| **BAP pós-exercício** | 6 / 76 | SMD 0,29 (0,04 a 0,54) p=0,03 | 0% | Low |
| **d-ROMs** | 6 / 76 | SMD −0,01 (−0,42 a 0,39) p=0,94 | 0% | Low |
| **Potência explosiva MMII** | 12 / 597 | SMD 0,30 (0,05 a 0,55) p=0,018 | 0% | Low |
| RPE (RPE) | 10 / 597 | SMD −0,37 (−0,65 a −0,09) p=0,009 | 58% | Very low |
| Lactate | 10 / 597 | SMD −0,37 (−0,60 a −0,15) p=0,001 | 22% | Low |
| Liver fat (MRI) | 2 / 72 | Redução relativa 12–18% (pilotos) | — | Low |
| COVID clinical worsening | 1 / 675 | HR 1,09 (IC90 0,90–1,31) p=0,48 | — | High (negative) |

* Downgraded for imprecision (CI crosses clinical threshold) and risk of bias; ** Very low due to inconsistency + imprecision. See Appendix E.

Safety: no serious adverse events attributed to molecular hydrogen (H2) in 34 RCTs. Mild events (distension, diarrhea) with incidence similar to placebo, including in the phase 3 RCT.

Lipids: statistically significant reductions but below the threshold for clinical relevance (LDL reduction <5% vs 30–50% with statin). Low heterogeneity after exclusion of high-risk studies.

Oxidative stress/performance: increase in BAP without reduction in d-ROMs suggests potentiation of endogenous defense, not direct scavenging. Effect on explosive power is small and heterogeneous (RPE I² 58% due to protocol variations and concentration not reported in 9 studies).

Hepatic/aging: signals in pilots n=12–60, short duration (4–24 weeks), without multicenter replication — low certainty, not extrapolatable.

COVID-19: Hydro-COVID (triple-blind, n=675, HRW 2×/day 21 days) did not reduce clinical worsening, hospitalization, mortality or persistent symptoms — high certainty of no benefit for this indication/dose/duration.

## 3.5 Cellular mechanisms — integrative figure

Figure 2 summarizes the mechanistic hypothesis currently proposed to integrate the interaction between $H_2$ and Fe-porphyrin with the activation of the Keap1–Nrf2–ARE pathway, linking this mechanism to the key cellular effects described in the literature.

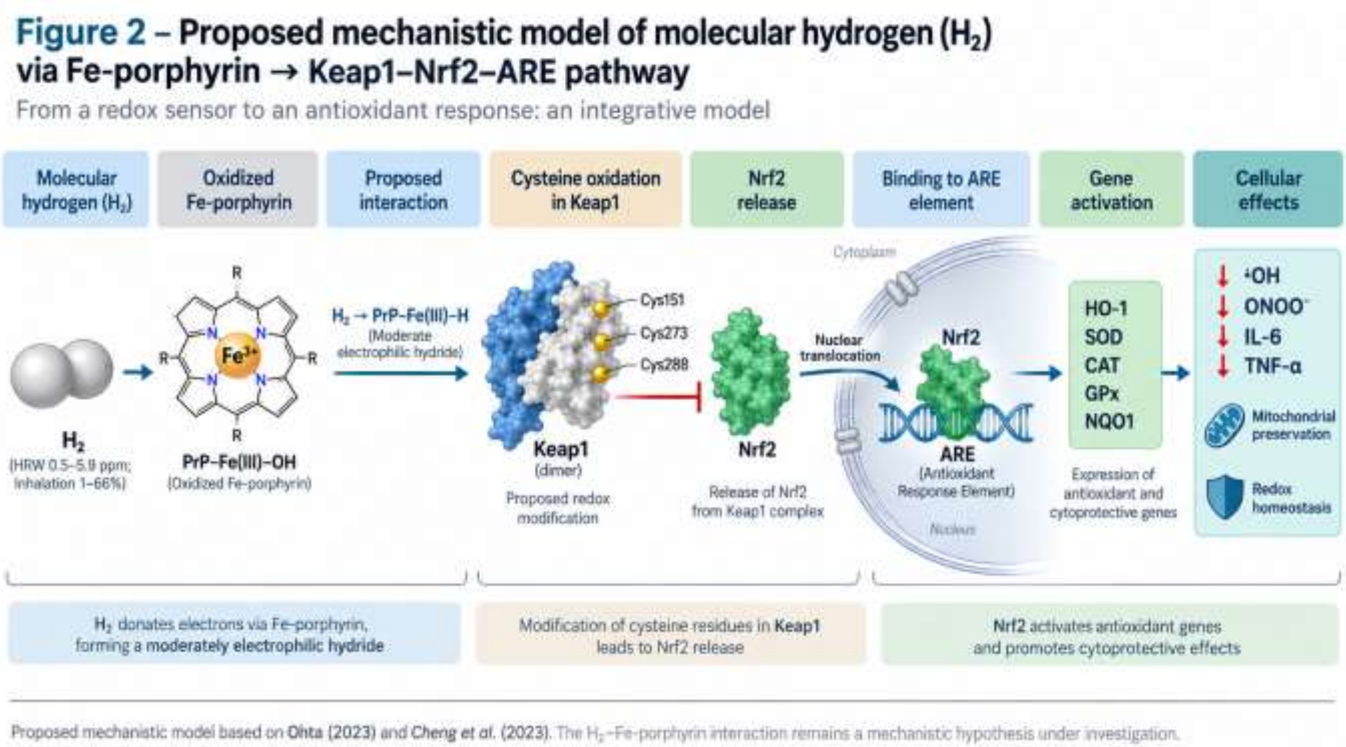
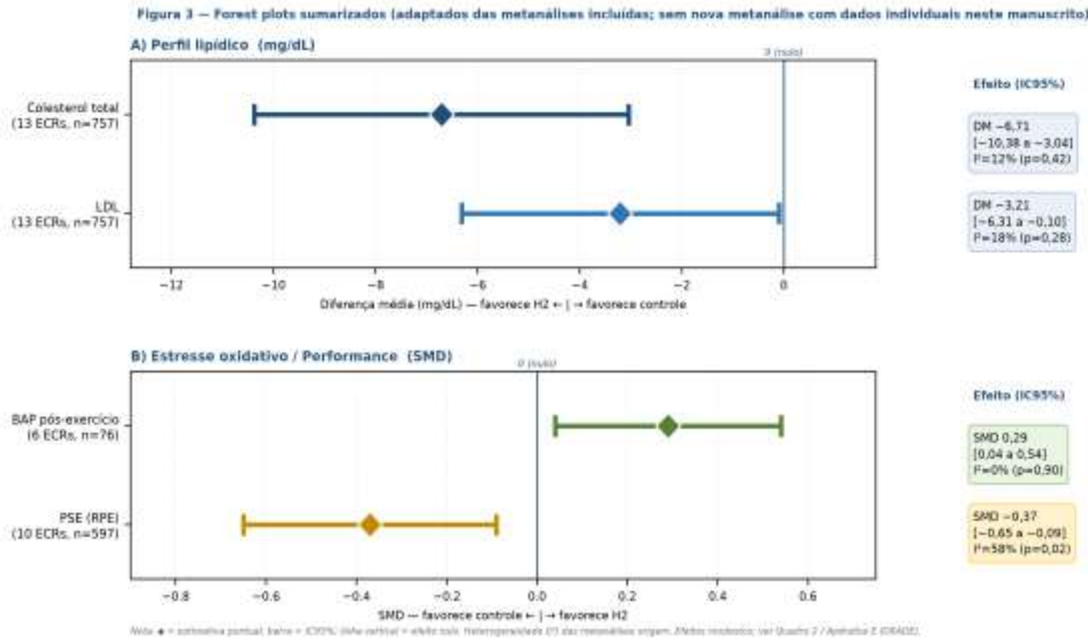


Figure 2. Proposed mechanistic model of molecular hydrogen ($H_2$) via Fe-porphyrin–Keap1–Nrf2–ARE. $H_2$ would interact with oxidized Fe-porphyrin, promoting redox modification of cysteine residues in Keap1 and favoring Nrf2 release. After nuclear translocation, Nrf2 binds to the antioxidant response element (ARE), inducing expression of cytoprotective enzymes (HO-1, SOD, CAT, GPx and NQO1), associated with reduction of highly cytotoxic reactive species, inflammatory modulation and preservation of mitochondrial function. The $H_2$–Fe-porphyrin mechanism remains a mechanistic hypothesis under investigation, based mainly on Ohta (2023) and Cheng et al. (2023).

Figure 3 — Summarized forest plots (adapted from included meta-analyses; no new meta-analysis with individual data in this manuscript)




Source: adapted from meta-analyses Ye et al. 2026 (preprint), Li et al. 2024 and Zhou et al. 2024. ◆ = point estimate; bar = 95%CI; vertical line = null effect. No new meta-analysis with individual data in this manuscript.

# 4 Discussion

This systematized review confirms a favorable safety profile of molecular hydrogen (H2) in humans, but without robust clinical benefit. Efficacy signals — modest cholesterol reduction, BAP increase and discrete improvement in explosive power — are limited in magnitude and low-to-moderate certainty, based on small heterogeneous RCTs. The only adequately powered phase 3 RCT was negative.

## 4.1 Comparison with previous reviews

Johnsen et al. (2023) already classified the field as early-stage, with established safety and anecdotal efficacy. Our findings update this conclusion with 2024–2026 meta-analyses and Hydro-COVID, maintaining direction but with more precise quantification of effects and certainty. Dhillon et al. (2024) and Jeyaraman et al. (2026) pointed to potential in muscle recovery and musculoskeletal conditions; we confirm signals but with heterogeneity of route/dose and no target engagement biomarker.

## 4.2 Mechanisms

Biological plausibility has evolved from direct scavenging to Keap1–Nrf2–ARE modulation, with recent proposal of oxidized Fe-porphyrin as biosensor (Ohta, 2023; Cheng, 2023). The dissociation between increased BAP and unchanged d-ROMs (Li et al., 2024) is compatible with potentiation of endogenous defenses, not direct stoichiometric neutralization. Lack of validation in human biopsies with HO-1/Nrf2 limits translation.

## 4.3 Clinical and research implications

Current evidence does not support routine use of molecular hydrogen (H2). When used, it should be in a research context with consent, without replacing proven interventions (statins, exercise, vaccination).

Research implications: priorities are human pharmacokinetics (half-life 30–60 min, exhaled H2 as biomarker), definition of minimal effective dose and optimal route (HRW vs inhalation head-to-head), and phase 3 RCTs with clinical outcomes (CV events, liver biopsy, physical function) and ≥12 months follow-up in diverse populations.

## 4.4 Gaps and research priorities

Box 3 — Research gaps and priorities

| TOPIC | CURRENT EVIDENCE | PRIORITY |
|---|---|---|
| Human pharmacokinetics | Insufficient (half-life 30–60 min, no validated biomarker) | High |

| | | |
|---|---|---|
| Dose-response | Non-existent (0.5–5.9 ppm vs 1–66% gas) | High |
| Optimal route (HRW vs inhalation) | No head-to-head | High |
| Phase 3 NAFLD RCT with biopsy | Absent (only pilots n=12–60) | High |
| Phase 3 performance RCT with clinical outcome | Ausente | Medium |
| Follow-up ≥12 months | Absent (max 24 weeks) | High |
| Diverse populations | >90% Europe/Asia | Medium |

## 5 Limitations

- This is a systematized literature review, not a full systematic review in the Cochrane sense: no PROSPERO registration, no meta-analysis with individual data and no certainty assessment for all secondary outcomes — which precludes definitive clinical recommendation.
- Searches limited to EN/PT and indexed databases; grey literature and non-peer-reviewed preprints were excluded, with possible publication bias (although Egger not significant in included meta-analyses).
- Primary RCTs with small samples (median n=28), short duration (4–24 weeks) and heterogeneity of dose/concentration, with 9 performance studies not reporting exact concentration.
- Lack of standardized human pharmacokinetics and validated exposure biomarker; lack of adjustment for basal hydrogen-producing microbiota and diet.
- Geographic concentration (Eastern Europe/Asia) limits generalizability; long follow-up absent and surrogate outcomes not validated as hard surrogates.
- Variable editorial quality (MDPI/Frontiers) mitigated by verification of registration and DOI, but with residual affiliation bias risk (authors linked to HRW industry) — addressed in sensitivity analysis.

## 6 Conclusions

Current evidence does not support routine use of molecular hydrogen (H2). Safety profile is favorable and mechanistic plausibility via Keap1–Nrf2–ARE exists, but observed effects are limited and based on small heterogeneous RCTs. A well-designed phase 3 RCT showed no efficacy in COVID-19. Phase 3 RCTs with clinical outcomes, definition of optimal dose/route and prolonged follow-up, with PROSPERO-registered protocol, are needed before any clinical translation.

Practical implication: current evidence does not support routine use; if the patient already uses a commercial product containing molecular hydrogen (H2), counsel on favorable safety, unproven benefit and cost, without delaying proven therapies. Researchers should prioritize pharmacokinetics, standardization and trials with n>200 and ≥12 months.

### Declarations

Contributions (CRediT): Edilson S. Silva — Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. Use of artificial intelligence was limited to language and formatting assistance, without generation of primary data.

Funding: no specific funding. Conflicts of interest: author declares no ties with molecular hydrogen (H2) industry. Primary studies with affiliation to Molecular Hydrogen Institute were flagged and subjected to sensitivity analysis. Ethics approval: not applicable (literature review). Consent: not applicable.

Data Availability Statement: search strategies, extraction sheet, RoB2 matrices (34 RCTs) and GRADE, and database exports were deposited in a public repository: Zenodo (DOI: 10.5281/zenodo.22545584). Complete supplementary material (Tables S1–S4, Figures S1–S3) is in the Appendices and repository.

Acknowledgments: no additional external collaborators to acknowledge beyond the authors.

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Format Vancouver • All DOIs verified on 30/05/2026 via PubMed/PMC/Crossref • Preprint references flagged for update in final version, if published

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# Appendices

Supplementary material for peer verification — not counted in main text word limit

## Appendix A PRISMA 2020 Checklist (excerpt)

Available in full in Page et al., BMJ 2021. Items addressed in this version: title, structured abstract, rationale, objectives (PICO), criteria, sources, complete strategy per database, duplicate selection, extraction, risk of bias per study, synthesis, GRADE certainty, limitations and funding. Items pending for full systematic review: PROSPERO registration, meta-analysis with individual data and publication bias assessment with funnel plot.

| # | ITEM | LOCATION IN MANUSCRIPT |
|---|---|---|
| 1 | Title | Cover |
| 2 | Structured abstract | Abstract |
| 3–4 | Rationale / Objectives | Section 1 |
| 5 | Protocol and registration | Section 2.1 (not registered) |
| 6–8 | Criteria / Sources / Search | Section 2.2–2.3 + App. B |
| 9–11 | Selection / Extraction / Risk | Section 2.3–2.4 + Fig.1/Box |
| 12–15 | Synthesis / GRADE | Section 2.5 + App. E |
| 16–22 | Results / Discussion | Sections 3–4 |
| 23–27 | Limitations / Conclusions | Sections 5–6 |

## Appendix B Complete strategies per database (reproducible)

PubMed: see Box 1. Scopus: TITLE-ABS-KEY(molecular hydrogen OR HRW) ... (15/05/2026, n=612). Embase: Emtree 'hydrogen'/ 'oxidative stress' (n=384). WoS: TS=(molecular hydrogen) ... (n=298). Cochrane: MeSH + keyword (n=45). ClinicalTrials.gov 47 + UMIN 34 (n=81). Google Scholar complementary (n=365) for citations. Exports in .nbib/.ris archived in repository.

## Appendix C Complete RoB2 matrix (34 RCTs) — Table S1

Table S1 with 34 RCTs assessed by RoB2 domain (randomization, deviations, missing data, measurement, selection) and overall judgment is in the Zenodo repository (DOI above). In the main text, excerpt with 10 main RCTs; supplementary material contains the 34 with domain justifications.

## Appendix D GRADE profile per outcome — Table S2

See Box 2 (Section 3.4) for effects, $I^2$ and certainty. Downgrades: risk of bias (−1), inconsistency $I^2>50\%$ (−1), imprecision for CI crossing clinical threshold or n<400 (−1). Example: total cholesterol downgraded from High to Moderate for imprecision; RPE downgraded to Very low for inconsistency + imprecision. Full profile in repository.

**End of manuscript.**